PROJECT 2: You and Data Science

Welcome to my project big project — and I am highly passionate about my subject and anything related to SPACE! I am a part of another class, ASTR 310, where I use python for computing in astronomy, and I would like to use my knowledge from that class and concepts from STAT 310* to create this project. I found this dataset on Kaggle, and I will carve the project around analyzing the basic traits of the stars, and getting to know interesting insights from the dataset. Hope you love it and learn something exciting! (:)*(*) 310A, 310B

SECTION 1: DATASET

Loading the dataset in a dataframe

```
import pandas as pd
df = pd.read_csv("star_classification.")
```

	Unnamed: 0	Luminosity (L/Lo)	Radius (R/Ro)	Temperature (K)	Spectral Class	A_type
0	Red	16.04473	0.879142	13.62005	TQ0.29424	ATV
1	Dwarf	2992.49072	1969.52167968	2552.9700235	2553.2847196	AQ0a
2	Barnard's Star	0.003476	0.185776	0.223771	3700.3936436	MP4e
3	Polaris	122.001002	27.96129104	27.946913	6346.329915	F7Ib
4	Barnard's Star	0.002350	0.189566	0.192308	3130.402269	MP4e
...	...	...	...	...	...	...
240	Red Giant	7.450170	4.431510	0.996087	2774.140000	MRV
241	Hyper	292.393444	49997.000001	9.070001	20010.000000	BYE
242	Dwarfs	239.782508	6391.020100	5.790011	23003.940799	BQE
243	Alpha Centauri B	0.504390	0.858992	0.855011	5298.300000	KTV
244	Red Giant	0.108799	3.464025	0.236524	2913.000000	MRV

1000 rows × 8 columns

Number of rows

```
df.shape[0]
```

```
1000
```

Number of columns

```
df.shape[1]
```

```
8
```

Names of the columns:

```
df.columns.tolist()
```

```
['Unnamed: 0',
 'Luminosity (L/Lo)',
 'Radius (R/Ro)',
 'Temperature (K)',
 'Spectral Class',
 'A_type']
```

SECTION 2: EXPLORATORY DATA ANALYSIS

I will explore the following:
Mean: the average temperature, luminosity, size of the stars
Median: the center of the dataset
Mode: the most common Spectral Class in stars
Max and Min: the range of temperatures in the stars
Percentage: comparing different elements of the dataset
Unpublished/related: largest and smallest star by size, distance to the star

```
df["Temperature (K)"].mean()
```

```
4860.0847
```

The average temperature of the stars in the dataset is **4860.0 K**

Let's Compare:
Hottest temperature recorded on Earth = **329.85 K (56.7° C/134° F)**
Our temperature = 5778 K (which is much hotter the sun!)
We cannot even begin to imagine how hot stars are, and these are just surface temperatures! The core temperatures are exponentially hot!

```
df["Temperature (K)"].max()
```

```
39846.000000
```

If you are unsure, the luminosity of a celestial object is basically its power output (amount of energy radiated per second). Here we can see, the average luminosity of a star is **10645**. Now this is in regards with the sun, where Sun's luminosity is 1. Isn't the difference mind-boggling?!

```
df["Luminosity (L/Lo)"].mean()
```

```
10645.0000
```

The average size of a star is about **57 times** the Sun!

```
df["Temperature (K)"].median()
```

```
3319.227812
```

This shows the center of the dataset is **3319 K**, 10% above this temperature and 50% below.

```
df[df["Spectral Class"].mode()] #to select the first alphabet of the spectral class types for easier analysis.
df
```

	Unnamed: 0	Luminosity (L/Lo)	Radius (R/Ro)	Temperature (K)	Spectral Class	A_type
0	Red	16.04473	0.879142	13.62005	TQ0.29424	ATV
1	Dwarf	2992.49072	1969.52167968	2552.9700235	2553.2847196	AQ0a
2	Barnard's Star	0.003476	0.185776	0.223771	3700.3936436	MP4e
3	Polaris	122.001002	27.96129104	27.946913	6346.329915	F7Ib
4	Barnard's Star	0.002350	0.189566	0.192308	3130.402269	MP4e
...	...	...	...	...	...	...
240	Red Giant	7.450170	4.431510	0.996087	2774.140000	MRV
241	Hyper	292.393444	49997.000001	9.070001	20010.000000	BYE
242	Dwarfs	239.782508	6391.020100	5.790011	23003.940799	BQE
243	Alpha Centauri B	0.504390	0.858992	0.855011	5298.300000	KTV
244	Red Giant	0.108799	3.464025	0.236524	2913.000000	MRV

1000 rows × 7 columns

```
df["Spectral Class".mode()
```

```
0    M
dtype: object
```

```
df["Spectral Class".mode()[0]
```

```
'M'
```

This shows the most common spectral class in this dataset! A, B, and O accounts for **20%** of the stars in the dataset. They have the following properties:
A = White, **B** = Blue, **O** = Blue-ish
Color **white**

Our ☉ Sun is G Spectral class!

```
df[df["Temperature (K)"].max()]
```

```
Unnamed: 0                 Hypergiant
Luminosity (L/Lo)          1.000000e+6
Radius (R/Ro)              2.000000e+3
Temperature (K)            39846.000000
Spectral Class                         O
A_type                               BYE
Name: 239, dtype: object
```

```
df[df["Temperature (K)"].min()]
```

```
Unnamed: 0                 Red Dwarfs
Luminosity (L/Lo)          0.000080
Radius (R/Ro)              0.008400
Temperature (K)            1939.000000
Spectral Class                         M
A_type                               MRV
Name: 223, dtype: object
```

The range of the temperatures is **39284 K**.

Now we will see where is the temperature ranges do the stars lie. According to astronomical data, any star with a temperature above 10000 K is a hot star, and any star with a temperature below 2000 K is a cool star.

```
len(df[df["Temperature (K)" > 10000]) # 100 K and below
```

```
24.1
```

```
len(df[df["Temperature (K)" < 2000]) # 100 K and below
```

```
24.1
```

23% of the stars are hot stars, whereas there are no "cool" stars in the dataset.

```
df.sample(1), "Radius (R/Ro)"
```

	Name	Distance (pc)	Luminosity (L/Lo)	Radius (R/Ro)	Temperature (K)	Spectral Class	sp_class
404	Betelgeuse	642.9640977	120904.0312500	887.0078398	3504.2200177	M2Iab	M

```
df.sample(1), "Temperature (K)"
```

	Name	Distance (pc)	Luminosity (L/Lo)	Radius (R/Ro)	Temperature (K)	Spectral Class	sp_class
380	Wolf 359	7.692776	0.000101	0.096907	2779.1400	M6V	M

The largest star is **Betelgeuse**, in the constellation of Orion (for those star lists), and it will explode in a supernova in the next tens or hundreds of thousands of years! And, the smallest star is **Wolf 359**, in the constellation of Leo.

```
df.sample(1), "Distance (pc)"
```

	Name	Distance (pc)	Luminosity (L/Lo)	Radius (R/Ro)	Temperature (K)	Spectral Class	sp_class
1	Dwarf	2992.49072	1969.52167968	2552.9700235	2553.2847196	AQ0a	A

```
df.sample(1), "Spectral Class"
```

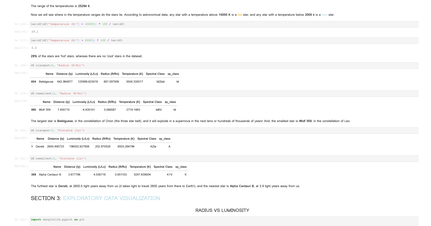
	Name	Distance (pc)	Luminosity (L/Lo)	Radius (R/Ro)	Temperature (K)	Spectral Class	sp_class
209	Alpha Centauri B	3.677748	4.258719	0.861103	5241.000000	KTV	K

The furthest star is **Dwarfs**, at 2553.3 light years away from us (it takes light to travel 2553 years from Rho to Earth), and the nearest star is **Alpha Centauri B**, at 3.3 light years away from us.

SECTION 3: EXPLORATORY DATA VISUALIZATION

RADIUS VS LUMINOSITY

```
import matplotlib.pyplot as plt
plt.style.use('dark_background')
plt.figure(figsize=(8, 6))
plt.scatter(df["Radius (R/Ro)"], df["Luminosity (L/Lo)"], color='orange', alpha=0.7)
plt.title('Radius vs Luminosity')
plt.xlabel('Luminosity (L/Lo)')
plt.ylabel('Radius (R/Ro)')
plt.grid(True, ls='-')
plt.show()
```



This plot demonstrates a correlation between a star's radius and its luminosity. Larger stars tend to be more luminous according to astronomy (stars with greater surface areas emit more energy), which is indicated by the graph clearly. We can also notice something else: most of the stars are clustered at the lower frequencies section (size similar to the Sun, and lower luminosities), which is predicted by astronomy (Main Sequence stars are the most abundant, with 90% of the stellar population in the universe). Also, according to the Stefan-Boltzmann Law, a star's luminosity is proportional to a power of its radius and temperature.

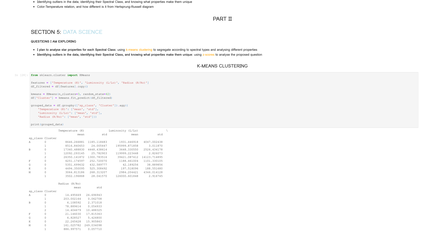
SPREAD OF TEMPERATURE BY SPECTRAL CLASS

```
plt.figure(figsize=(8, 6))
df.boxplot(column='Temperature (K)', by='Spectral Class', grid=False)
plt.suptitle('')
plt.title('Temperature Range by Spectral Class')
plt.xlabel('Spectral Class')
plt.ylabel('Temperature (K)')
plt.show()
# Temperature range in each class
```

The box and whisker that stellar temperature decreases across spectral classes from A to M, with hotter classes exhibiting a wider temperature range and cooler classes showing a narrower, lower range. Class B stars (10000 K to 30000 K) have the greatest spread, while classes F, G, K, and M are cooler and more consistent in temperature, with the data more clustered near their median. We can hence see that molecules tend to be more diverse in temperature.

DISTANCE DISTRIBUTION

```
plt.figure(figsize=(8, 6))
plt.hist(df["Distance (pc)"], bins=50, color='lightblue', alpha=0.75)
plt.title('Distance Distribution of Stars')
plt.xlabel('Distance')
plt.ylabel('Frequency')
plt.show() # Distance distribution of stars?
```



This histogram shows the distance distribution of stars in light-years (ly). Most stars in the dataset are relatively close, with the majority located within a few hundred light-years, particularly in the first ten bins. A smaller number of stars are farther away, but these are significantly distant, with only a few stars found beyond 600 light-years and one exceptionally beyond 2000 light-years. This distribution suggests that closer stars are more commonly observed, likely due to observational limits where distant stars are harder to detect.

SECTION 4: PLANNING FOR PART II

OBSERVATIONS MADE

- There are many correlations between the properties of stars
- The dataset did agree with the astronomical fundamentals of stars
- The data set around lower-median in mass stars, and has limited data for the outlying values (hotter/brighter, massier/far, smaller/larger, etc)
- There are no O class stars

PART II PLAN:

- I want to create a Hertzsprung-Russell diagram (luminosity vs temperature), which is an astronomical graph of stars, showing evolutionary stages or classes
- I plan to analyze star properties for each Spectral Class
- Identifying outliers in the data: identifying their Spectral Class, and knowing what properties make them unique
- Color Temperature relation, and how different is it from Hertzsprung-Russell diagram

PART II

SECTION 5: DATA SCIENCE

QUESTIONS I AM EXPLORING:

- I plan to analyze star properties for each Spectral Class, using K-means clustering, to segregate according to spectral types and analyzing different properties
- Identifying outliers in the data: identifying their Spectral Class, and knowing what properties make their unique using K-means to analyze the proposed question

K-MEANS CLUSTERING

```
from sklearn.cluster import KMeans

features = ['Temperature (K)', 'Luminosity (L/Lo)', 'Radius (R/Ro)']
X = df[features].dropna()
# Standardize features for better clustering
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
groupby_data = df.groupby('Spectral Class', 'Clusters').agg(
    'Temperature (K)': ['mean', 'std'],
    'Luminosity (L/Lo)': ['mean', 'std'],
    'Radius (R/Ro)': ['mean', 'std']
)
print(groupby_data)
```

```
                     Temperature (K)           Luminosity (L/Lo)
Clusters                        mean       std              mean       std
Spectral Class
A                       9520.000000  2200.000000      1001.000000  2000.000000
B                       2014.000000    20.000000      1000.000000    0.000000
F                       3014.000000    20.000000      1000.000000    0.000000
G                       2014.000000    20.000000      1000.000000    0.000000
K                       20000.000000  1000.000000     20000.000000  1000.000000
M                       20000.000000  1000.000000     20000.000000  1000.000000
Red Dwarfs              1939.000000     0.000000         0.000080     0.000000
Red Giant               3014.000000    20.000000      1000.000000    0.000000
Red Giants              3014.000000    20.000000      1000.000000    0.000000
Red Supergiant          3014.000000    20.000000      1000.000000    0.000000
Red Supergiants         3014.000000    20.000000      1000.000000    0.000000
Supergiant              3014.000000    20.000000      1000.000000    0.000000
Supergiants             3014.000000    20.000000      1000.000000    0.000000

                     Radius (R/Ro)
                              mean       std
Spectral Class
A                     1000.000000  1000.000000
B                        0.000000     0.000000
F                        0.000000     0.000000
G                        0.000000     0.000000
K                     1000.000000  1000.000000
M                     1000.000000  1000.000000
Red Dwarfs               0.008400     0.000000
Red Giant                0.000000     0.000000
Red Giants               0.000000     0.000000
Red Supergiant           0.000000     0.000000
Red Supergiants          0.000000     0.000000
Supergiant               0.000000     0.000000
Supergiants              0.000000     0.000000
```

ANALYSIS

Spectral Class A

- Cluster 0: The majority of A-class stars have moderate temperatures (~9999 K), high variability in luminosity (mean = 1001.40, std = 4041.00), and relatively small radii (mean = 16.50). This suggests that these are relatively typical main-sequence stars for this class.
- Cluster 1: These stars are exceptionally luminous (mean = 196,999.67) and large (radius mean = 20.00), with almost no variability in their properties (very low std). These likely represent supergiant stars within the A class.

Spectral Class B

- Cluster 0: B-class stars in this cluster are moderately hot (mean temperature ~17,242 K) and luminous (mean luminosity ~3969.33) but relatively small (radius ~8.12). These are likely massive main-sequence stars.
- Cluster 1: These stars are less hot (mean ~12,092 K) but significantly more luminous (mean luminosity ~119,999.25) and larger (radius mean ~79.89). They are likely B-class giants or supergiants.

Spectral Class F

- Cluster 0: F-class stars show a mean temperature of ~5931 K, relatively low luminosity (mean = 42.16), and small radii (mean = 8.83). These represent typical G-class main-sequence stars, such as our Sun.

Spectral Class G

- Cluster 0: G-class stars have slightly cooler temperatures (~4999 K), moderate luminosity (~197.52), and moderate radii (~22.27). These are likely giant stars within this spectral class.

Spectral Class M

- Cluster 0: M-class stars in this cluster are relatively cool (~3994 K) with moderate luminosity (mean = 2994.25) and high variability in radius (mean = 161.03, std = 249.05). The variability suggests a mix of smaller main-sequence stars and larger red giants.
- Cluster 1: These stars are highly luminous (mean luminosity = 126,000.00) with massive radii (~980.00) and almost no variability. These are likely red supergiants.

OUTLIERS

```
mask_id = df["Temperature (K)"] > mean_temp[df["Temperature (K)"]] & df["Luminosity (L/Lo)"] > mean_lum[df["Temperature (K)"]]
df["Temp_outlier"] = mask_id["Temperature (K)"]
df["Lum_outlier"] = mask_id["Luminosity (L/Lo)"]
df["Radius_outlier"] = df["Radius (R/Ro)"] > mean_rad[df["Temperature (K)"]] & df["Luminosity (L/Lo)"] > mean_lum[df["Temperature (K)"]]
print("Temperature outliers:", df["Temp_outlier"].sum())
print("Luminosity outliers:", df["Lum_outlier"].sum())
print("Radius outliers:", df["Radius_outlier"].sum())
```

```
Temperature outliers: 0
Luminosity outliers: 0
Radius outliers: 0
```

```
outliers = df[
    (df["Temp_outlier"] abs) & 0) |
    (df["Lum_outlier"] abs) & 0) |
    (df["Radius_outlier"] abs) & 0)
]
print("Total outliers:", len(outliers))
outliers[['Name', 'Spectral Class', 'Temperature (K)', 'Luminosity (L/Lo)', 'Radius (R/Ro)', 'Distance (pc)']].drop_duplicates(subset=['Name'])
```

```
Total outliers: 0
```

	Name	Spectral Class	Temperature (K)	Luminosity (L/Lo)	Radius (R/Ro)	Distance (pc)
0	Red Dwarfs	Red Dwarfs	1939.000000	0.000080	0.008400	2553.000000
1	Red Giant	Red Giant	3014.000000	1000.000000	0.000000	2553.000000
2	Red Giants	Red Giants	3014.000000	1000.000000	0.000000	2553.000000
3	Red Supergiant	Red Supergiant	3014.000000	1000.000000	0.000000	2553.000000
4	Red Supergiants	Red Supergiants	3014.000000	1000.000000	0.000000	2553.000000
5	Supergiant	Supergiant	3014.000000	1000.000000	0.000000	2553.000000
6	Supergiants	Supergiants	3014.000000	1000.000000	0.000000	2553.000000

Antennae Galaxies

```
df[df["Spectral Class"] & df["Temperature (K)"] .apply(lambda x: abs(x[0]
spectral_class_mean = df.groupby('Spectral Class', 'Temperature (K)', 'Luminosity (L/Lo)', 'Radius (R/Ro)', 'Distance (pc)').mean()
print("Spectral Class Mean Values:")
print(spectral_class_mean)
```

```
Spectral Class Mean Values:
                                                
```